Sahaj Bhadja

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Summary

Software engineer experienced in **JavaScript, React, Python, PostgreSQL, and MySQL**, shipping customer-facing AI product features end-to-end and optimizing applications for speed, reliability, and scale.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Architected a multi-tenant AI education platform on Azure SQL and PostgreSQL with normalized schemas for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through query optimization.
- Restructured a fault-tolerant Python FastAPI LLM orchestration system processing 1,000+ AI jobs with explicit job-state transitions, increasing concurrent throughput 3x while maintaining 99%+ reliability at production scale.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Designed React dashboards and Python Flask backends for an AI grading product serving 150+ instructors, processing 10,000+ daily requests with MySQL persistence and stable API performance at scale.
- Integrated OpenAI GPT-4o APIs with Claude and vision models to evaluate 10,000+ submissions, adding reusable fallbacks and automated tests while reducing customer feedback turnaround time 60% at scale.

USC AD Labs **Los Angeles**
ML Research Assistant *January 2025 – May 2025*
- Built a GPT-based Decision Transformer for offline reinforcement learning in autonomous driving, improving long-horizon trajectory planning 18% in simulation while processing 100,000+ trajectory logs end to end.
- Architected a spatial-temporal encoder integrating HD-map geometry covering road layout and lane direction with ego-agent past trajectories via multi-head attention, improving closed-loop planning stability 15% at scale.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a React and Node.js resource management platform with REST APIs and microservices, syncing two million customer records while reducing resource wastage 42% and increasing conversions 15% at scale.
- Shipped JavaScript and React product features in a fast-paced environment, using Google Analytics and Sentry monitoring to improve user experience while reducing deployment time 50% through Jenkins CI/CD.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Analyzed client requirements, mapped 12+ integration points, and presented architecture recommendations before implementing React Native and React booking, payment, and real-time status experiences for end users at scale.
- Delivered Python Flask and Firebase solutions across 50,000+ records, validating implementation against user needs while achieving sub-200-millisecond responses and reducing manual refund handling 70% at production scale.

Skills

Languages: JavaScript, TypeScript, Python, Java, SQL
Frontend: React, Next.js, HTML, CSS
Backend: Flask, FastAPI, Node.js, Express.js, REST APIs
Data: PostgreSQL, MySQL, MongoDB
AI/ML: OpenAI API, PyTorch, Transformers, LLM Systems, NLP
Delivery: CI/CD, Performance Optimization, Docker, Kubernetes, Git, AWS, Azure, Agile

Projects

Chat2SQL Code | *JavaScript, PostgreSQL, MySQL*
- Built an AI-powered JavaScript and TypeScript SQL workspace supporting PostgreSQL, MySQL, and 10+ databases, accelerating query authoring, schema exploration, and reliable Docker-based deployment from a unified interface at scale.

Big-Brain Live | *Customer-Facing AI Product*
- Built a customer-facing AI document platform with PostgreSQL, LLaMA-2, and FAISS, enabling conversational search and summarization for 200+ users while reducing document-analysis time 60% with sub-second retrieval at scale.

PiXMod Live | *JavaScript Product Experience*
- Designed a Next.js JavaScript SaaS image platform using CLIP, SAM, and Stable Diffusion, processing 1,000+ edited assets with Stripe subscriptions and MongoDB storage for end users at scale.